

MATH1141: Higher Mathematics 1A (Calculus)

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Contents

0.1	Content Overview	3
1	Sets, inequalities and functions	4
1.1	Sets of numbers	4
1.1.1	Intervals	4
1.1.2	Solving Inequalities	4
1.1.3	Absolute Values	5
1.1.4	Functions	6
1.1.5	The Elementary Functions	7
1.1.6	Implicitly Defined Functions	7
1.1.7	Continuous Functions	7
2	Limits	9
2.1	Limits of functions at infinity	9
2.1.1	Basic rules for limits	9
2.1.2	The pinching theorem	10
2.1.3	Limits of the form $f(x)/g(x)$	11
2.1.4	Limits of the form $\sqrt{f(x)} - \sqrt{g(x)}$	12
2.1.5	Indeterminate forms	12
2.2	The definition of $\lim_{x \rightarrow \infty} f(x)$	12
2.3	Proving that $\lim_{x \rightarrow \infty} f(x) = L$ using the limit definition	13
2.4	Proofs of basic limit results	13
2.5	Limits of functions at a point	14
2.5.1	Left-hand, right-hand and two-sided limits	14
2.5.2	Limits and continuous functions	15
2.5.3	Rules for limits at a point	15
3	Properties of continuous functions	18
3.1	Combining continuous functions	18
3.2	Continuity on intervals	18
3.3	The intermediate value theorem	19
3.4	The maximum-minimum theorem	19
4	Differentiable functions	21
4.1	Gradients of tangents and derivatives	21
4.2	Rules for differentiation	22
4.3	Proofs of results	23
4.4	Implicit differentiation	23
4.5	Differentiation, continuity and split functions	23
4.6	Derivatives and function approximation	24
4.7	Derivatives and rates of change	24
4.8	Local maximum, local minimum and stationary points	24

5	The mean value theorem and applications	25
5.1	The mean value theorem	25
5.2	Proof of the mean value theorem	26
5.3	Proving inequalities using the mean value theorem	27
5.4	Error bounds	28

0.1 Content Overview

- Functions: continuity, differentiability, invertibility (inverse functions)
- Curve sketching
- Integration: area, Riemann sum (approximation), Fundamental theorems of calculus
- Logarithm: exponential, hyperbolic

Chapter 1

Sets, inequalities and functions

1.1 Sets of numbers

A set is a collection of distinct objects. The objects in a set are elements or members of the set. Commonly used sets:

- The set of natural numbers: $\mathbb{N} = \{0, 1, 2, 3, 4, \dots\}$
- The set of integers: $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$
- The set of rational numbers: $\mathbb{Q} = \{\frac{p}{q} : p, q \text{ are integers and } q \neq 0\}$
- The set of $\mathbb{R}$ real numbers may be represented as the collection of points lying on the number line

If A is a set of numbers and the number x is a member of the set A , then we write:

$$x \in A$$

If x is not a member of A , then we write

$$x \notin A$$

1.1.1 Intervals

- Parenthesis: excludes endpoints
- Bracket: includes endpoints
- Combination: neither open nor closed interval

1.1.2 Solving Inequalities

For $x, y, z \in \mathbb{R}$:

- if $x > y$, then $x + z > y + z$
- if $x > y$ and $z > 0$, then $xz > yz$
- if $x > y$ and $z < 0$, then $xz < yz$

Example

Solve the quadratic inequality $x^2 + 4x > 21$

$$\begin{aligned}
 x^2 + 4x > 21 &\iff x^2 + 4x - 21 > 0 \\
 &\iff (x + 7)(x - 3) > 0
 \end{aligned}$$

Solution set: $x < -7$ or $x > 3$

$$x \in (-\infty, -7) \cup (3, \infty)$$

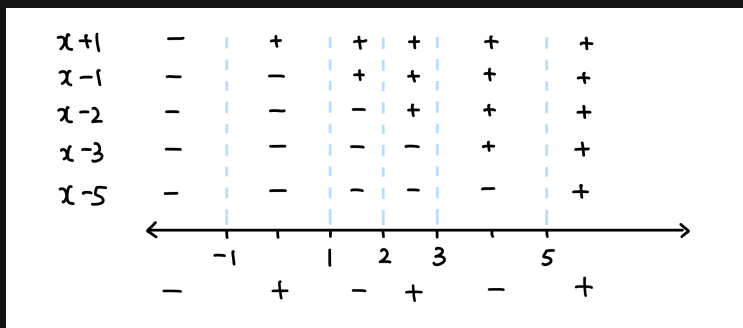
Example

Solve the rational inequality $\frac{1}{x+1} < \frac{1}{(x-2)(x-3)}$

Multiply both sides by $(x+1)^2(x-2)^2(x-3)^2$ ($>= 0$),

$$\begin{aligned}
 \frac{1}{x+1} < \frac{1}{(x-2)(x-3)} &\iff (x+1)(x-2)^2(x-3)^2 < (x+1)^2(x-2)(x-3) \\
 &\iff (x+1)(x-2)^2(x-3)^2 - (x+1)^2(x-2)(x-3) < 0 \\
 &\iff (x+1)(x-2)(x-3)(x-4)(x-5) < 0
 \end{aligned}$$

Solution set: $(-\infty, -1) \cup (1, 2) \cup (3, 5)$



1.1.3 Absolute Values

The absolute value (or magnitude) of a real number x is

$$|x| = \begin{cases} x & \text{if } x \geq 0, \\ -x & \text{if } x < 0 \end{cases}$$

Example

Solve the inequality $|3x + 1| \leq 4$

$$\begin{aligned}
 |3x + 1| \leq 4 &\iff \left|x + \frac{1}{3}\right| \leq \frac{4}{3} \\
 &\iff x + \frac{1}{3} \leq \frac{-4}{3} \quad \text{or} \quad x + \frac{1}{3} \leq \frac{4}{3} \\
 &\iff x \leq \frac{-5}{3} \quad \text{or} \quad x \leq 1
 \end{aligned}$$

Solution set: $(-\infty, \frac{-5}{3}] \cup [1, \infty)$

Example

Solve the inequality $\frac{|x+5|}{|x-11|} < 1$

$$\frac{|x+5|}{|x-11|} < 1 \iff |x+5| < |x-11| \quad (\text{multiply both sides by } |x-11|)$$

$$\iff |x+5|^2 < |x-11|^2$$

$$\iff x^2 + 10x + 25 < x^2 - 22x + 121$$

$$\iff 32x < 96$$

$$\iff x < 3$$

Solution set: $(-\infty, 3)$

Note: 11 is not in the solution set

1.1.4 Functions

Domain, codomain and range

Let A and B be subsets of $\mathbb{R}$: A function with domain A and codomain B is a rule which assigns to every $x \in A$ exactly one number $f(x) \in B$

$$f : A \rightarrow B$$

$$f : A \ni x \mapsto f(x) \in B$$

- $A = \text{Dom}(f)$: set of all inputs
- $B = \text{Codom}(f)$: set that contains all the outputs
- Range of f : set of actual outputs, defined by:

$$\text{Range}(f) = \{f(x) \in B : x \in A\}$$

- Hence, $\text{Range}(f) \subseteq \text{Codom}(f)$
- The **codomain does not have to be unique**
- The codomain is useful when the range is unknown

Example

Given: $f : [0, \infty) \rightarrow \mathbb{R}, f(x) = 2 + \sqrt{x}, \forall x \in [0, \infty)$

Then, $\text{Dom}(f) = [0, \infty)$, $\text{Codom}(f) = \mathbb{R}$, and $\text{Range}(f) = [2, \infty)$

Natural domain (or maximal domain) of f is the largest possible domain for which the rule makes sense (for real numbers)

Forming new functions

Let A, B, C and D be subsets of $\mathbb{R}$. Given two functions $f : A \rightarrow B$ and $g : A \rightarrow B$, then the functions

$$f + g, \quad f - g, \quad fg, \quad \frac{f}{g}$$

are defined by the rules

$$\begin{aligned}(f + g)(x) &= f(x) + g(x), \quad \forall x \in A \\(f - g)(x) &= f(x) - g(x), \quad \forall x \in A \\(fg)(x) &= f(x)g(x), \quad \forall x \in A \\ \left(\frac{f}{g}\right)(x) &= \frac{f(x)}{g(x)}, \quad \forall x \in A \text{ provided that } g(x) \neq 0.\end{aligned}$$

To form these new functions, the domains of both f and g must be the same

Suppose that $f : C \rightarrow D$ and $g : A \rightarrow B$ are functions such that

$$\text{Range}(g) \subseteq \text{Dom}(f) = C$$

Then the composition function $(f \circ g)(x) = f(g(x)), \quad \forall x \in A$

1.1.5 The Elementary Functions

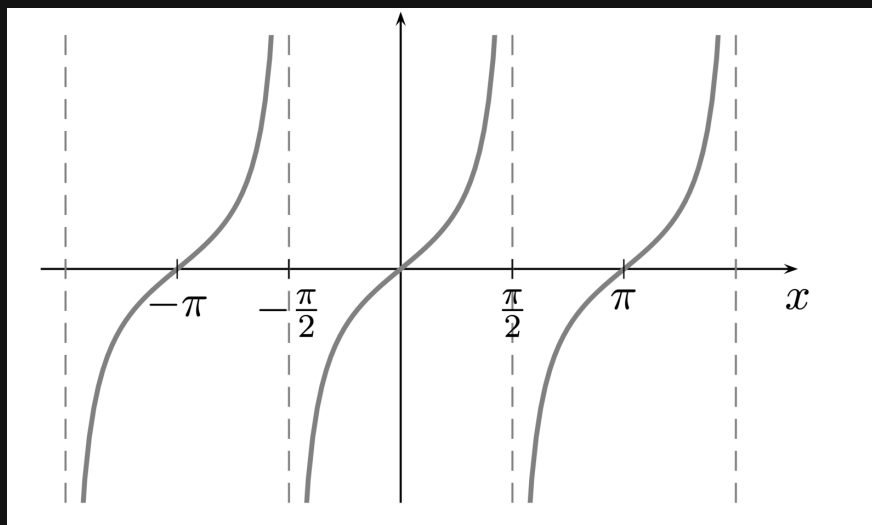
Elementary functions are those that can be constructed by combining a finite number of polynomials, exponentials, logarithms, roots, trigonometric and inverse trigonometric functions via function composition ($\circ$), $+$, $-$, $\times$, and $\div$.

1.1.6 Implicitly Defined Functions

Many curves on the plane can be described by the points (x, y) that satisfy some equation involving x and y . However, not all these curves are necessarily functions. Some cannot be expressed with a single y term ($y = \dots$). In this case, the curve may be decomposed into two or more functions that are implicitly defined by the curve.

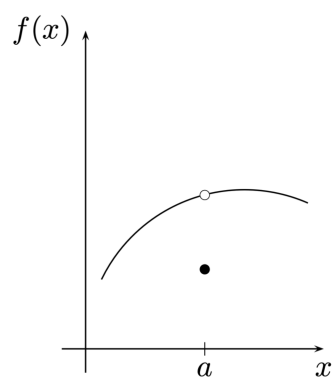
1.1.7 Continuous Functions

In the below example, the $\tan x$ graph breaks at $x = -\frac{\pi}{2}, \frac{\pi}{2}, \dots$

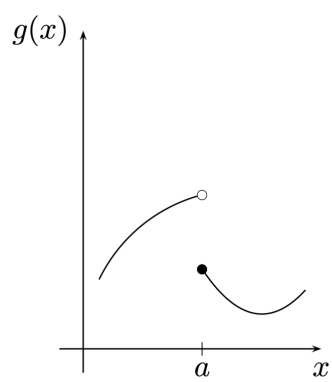


However, this break in the domain does not make it a discontinuity in the graph. Discontinuities can be categorised as following:

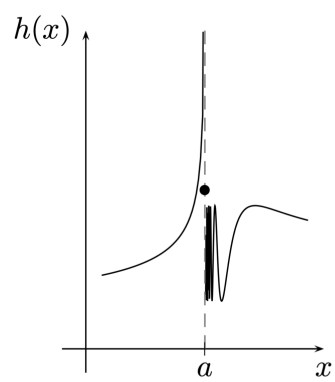
removable



jump



essential



Chapter 2

Limits

2.1 Limits of functions at infinity

- If $f(x)$ gets arbitrarily close to some $L \in \mathbb{R}$ as x tends to ∞ , then we write:

$$\lim_{x \rightarrow \infty} f(x) = L \quad \text{or} \quad f(x) \rightarrow L \text{ as } x \rightarrow \infty$$

- If $f(x)$ gets arbitrarily large as x tends to ∞ , then we write:

$$f(x) \rightarrow \infty \text{ as } x \rightarrow \infty$$

- In this case, we state that the limit of $f(x)$ as x tends to ∞ does not exist

$$\lim_{x \rightarrow \infty} \frac{1}{f(x)} = 0$$

- The limit notation can only be used when the limit has a constant definition (do not write $\lim_{x \rightarrow \infty} = \infty$)

2.1.1 Basic rules for limits

If $\lim_{x \rightarrow \infty} f(x) = L_1$ and $\lim_{x \rightarrow \infty} g(x) = L_2$ then

- $\lim_{x \rightarrow \infty} (f(x) + g(x)) = L_1 + L_2$
- $\lim_{x \rightarrow \infty} (f(x) - g(x)) = L_1 - L_2$
- $\lim_{x \rightarrow \infty} f(x)g(x) = L_1L_2$
- $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \frac{L_1}{L_2}$, provided that $L_2 \neq 0$

Example

Find the limit of $f(x) = \frac{1 + \frac{2}{3x+4}}{5 - 6e^{-x}}$ as $x \rightarrow \infty$ (if it exists)

$$\begin{aligned}\lim_{x \rightarrow \infty} f(x) &= \frac{\lim_{x \rightarrow \infty} (1 + \frac{2}{3x+4})}{\lim_{x \rightarrow \infty} (5 - 6e^{-x})} \\&= \frac{\lim_{x \rightarrow \infty} 1 + \lim_{x \rightarrow \infty} \frac{2}{3x+4}}{\lim_{x \rightarrow \infty} 5 - \lim_{x \rightarrow \infty} 6e^{-x}} \\&= \frac{1+0}{5-0} \\&= \frac{1}{5}\end{aligned}$$

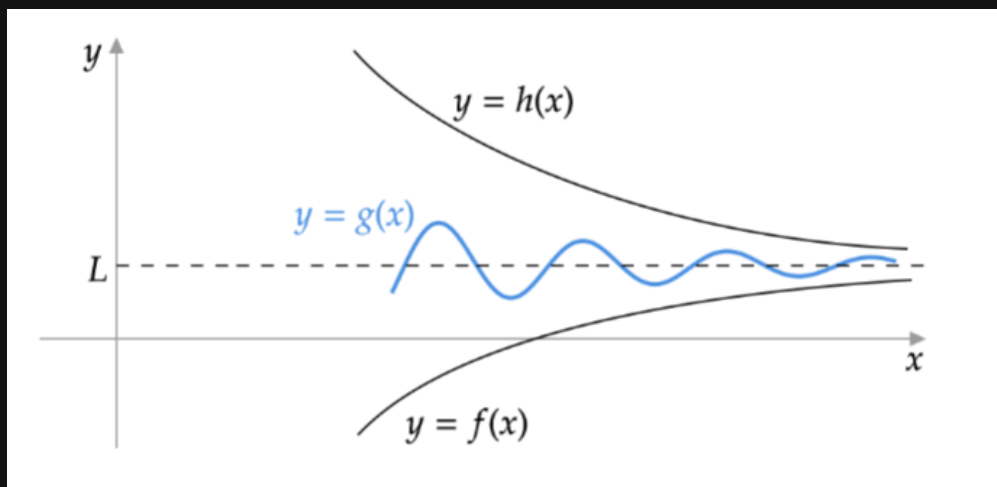
2.1.2 The pinching theorem

Suppose that f and h have the same limit as $x \rightarrow \infty$. If the graph of g always lies between the graphs of f and h , then g has the same limit.

Theorem Let f , g , and h be functions defined on (b, ∞) for some $b \in \mathbb{R}$. If :

- $f(x) \leq g(x) \leq h(x)$, $\forall x \in (b, \infty)$, and
- $\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} h(x) = L$, then

$$\lim_{x \rightarrow \infty} g(x) = L$$



Example

Find the limit of $g(x) = e^{-x^2} 2^{\sin 3x}$ as $x \rightarrow \infty$ (if it exists).

$$\sin x \in [-1, 1], \quad \forall x \in \mathbb{R}$$

$$\Rightarrow \frac{1}{2} \leq 2^{\sin 3x} \leq 2, \quad \forall x \in \mathbb{R}$$

$$\Rightarrow \frac{e^{-x^2}}{2} \leq e^{x^2} 2^{\sin 3x} \leq 2e^{-x^2}$$

$$\text{Since } \lim_{x \rightarrow \infty} \frac{e^{-x^2}}{2} = \lim_{x \rightarrow \infty} 2e^{-x^2} = 0,$$

$$\therefore \lim_{x \rightarrow \infty} e^{-x^2} 2^{\sin 3x} = 0$$

2.1.3 Limits of the form $f(x)/g(x)$

When calculating $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)}$ when both $f(x)$ and $g(x)$ tend to ∞ as $x \rightarrow \infty$, the previous rules cannot be applied. The main goal in this situation is to **divide both f and g by the fastest growing term in g** .

Example

Find the limit $\lim_{x \rightarrow \infty} \frac{5x^3 + 6x^2 - 4 \sin x}{\cos 3x + 5x - 2x^3}$ (if it exists).

$$\begin{aligned} \frac{5x^3 + 6x^2 - 4 \sin x}{\cos 3x + 5x - 2x^3} &= \frac{5 + \frac{6}{x} - 4 \frac{\sin x}{x^3}}{\frac{\cos 3x}{x^3} + \frac{5}{x^2} - 2} \\ &= \frac{5 + 0 + 0}{0 + 0 - 2} \\ &= -\frac{5}{2} \quad \text{as } x \rightarrow \infty \end{aligned}$$

Example

Find the limit $\lim_{x \rightarrow \infty} \frac{x^2 + 3x}{\sqrt{2x^4 + 3} - 4x}$ (if it exists).

$$\begin{aligned} \frac{x^2 + 3x}{\sqrt{2x^4 + 3} - 4x} &= \frac{1 + \frac{3}{x}}{\frac{1}{x^2} \sqrt{2x^4 + 3} - \frac{4}{x}} \\ &= \frac{1 + \frac{3}{x}}{\sqrt{2 + \frac{3}{x^4}} - \frac{4}{x}} \\ &\rightarrow \frac{1 + 0}{\sqrt{2 + 0} - 0} \\ &= \frac{1}{\sqrt{2}} \quad \text{as } x \rightarrow \infty \end{aligned}$$

2.1.4 Limits of the form $\sqrt{f(x)} - \sqrt{g(x)}$

In this case, the leading term may not help. Instead the lower order term will usually determine the limit. This can be achieved by **multiplying the numerator and denominator by the conjugate**.

Example

Find the limit of $\lim_{x \rightarrow \infty} \sqrt{2x^2 + 3x} - \sqrt{2x^2 - x}$.

$$\begin{aligned} \sqrt{2x^2 + 3x} - \sqrt{2x^2 - x} &= \frac{(\sqrt{2x^2 + 3x} - \sqrt{2x^2 - x})(\sqrt{2x^2 + 3x} + \sqrt{2x^2 - x})}{(\sqrt{2x^2 + 3x} + \sqrt{2x^2 - x})} \\ &= \frac{2x^2 + 3x - (2x^2 - x)}{\sqrt{2x^2 + 3x} + \sqrt{2x^2 - x}} \\ &= \frac{4x}{\sqrt{2x^2 + 3x} + \sqrt{2x^2 - x}} \\ &= \frac{x}{\sqrt{2 + \frac{3}{x}} + \sqrt{2 - \frac{1}{x}}} \\ &\Rightarrow \frac{4}{\sqrt{2} + \sqrt{2}} \\ &= \sqrt{2} \quad \text{as } x \rightarrow \infty \end{aligned}$$

2.1.5 Indeterminate forms

Some limits such as the form $\frac{\infty}{\infty}$ cannot be determined in advance and is hence called an indeterminate form. Other indeterminate forms include:

- $\infty - \infty$ if $f(x) \rightarrow \infty$ and $g(x) \rightarrow \infty$ as $x \rightarrow \infty$, $f(x) - g(x) \rightarrow ?$
- $\frac{0}{0}$ if $f(x) \rightarrow 0$ and $g(x) \rightarrow 0$ as $x \rightarrow \infty$, $\frac{f(x)}{g(x)} \rightarrow ?$
- $0 \times \infty$ if $f(x) \rightarrow 0$ and $g(x) \rightarrow \infty$ as $x \rightarrow \infty$, $f(x)g(x) \rightarrow ?$

2.2 The definition of $\lim_{x \rightarrow \infty} f(x)$

Suppose that $b, L \in \mathbb{R}$, and f is a real-valued function defined on (b, ∞) . It is said:

$$\lim_{x \rightarrow \infty} f(x) = L$$

if for every $\epsilon > 0$, there is $M \in \mathbb{R}$ such that:

$$x > M \Rightarrow |f(x) - L| < \epsilon$$

2.3 Proving that $\lim_{x \rightarrow \infty} f(x) = L$ using the limit definition

Example

Prove that $\lim_{x \rightarrow \infty} \frac{5x}{x+3} = 3$.

$|f(x) - L| < \epsilon$ if ?

$$\begin{aligned} |f(x) - L| &= \left| \frac{5x}{x+3} - 3 \right| = \left| \frac{5x - 3(x+3)}{x+3} \right| = \left| \frac{-15}{x+3} \right| \\ &= \frac{15}{x+3} \quad \text{for } x > 3 \\ &< \frac{15}{x} \quad \text{for } x > 0 \end{aligned}$$

$$\Rightarrow |f(x) - L| < \epsilon \quad \text{if} \quad \frac{15}{x} < \epsilon \text{ and } x > 0$$

$$\text{(i.e. } x > \frac{15}{\epsilon} > 0 \text{)}$$

$$\Rightarrow \text{For any } \epsilon > 0, \exists M = \frac{15}{\epsilon} \text{ such that}$$

$$|f(x) - L| < \epsilon \text{ whenever } x > M$$

Strategy for finding M

1. Find a good upper bound for $|f(x) - L|$
2. Find a simple condition on x such that this upper bound is less than ϵ
3. Use this condition to state an appropriate value for M

Note: The value of M is not unique. If M_0 is a value for M , then any upper bound of M_0 is also a value for M

Example

Show that $\lim_{x \rightarrow \infty} \frac{x^2 - \cos x}{x^2 + 1} = 1$

$$\begin{aligned} \left| \frac{x^2 - \cos x}{x^2 + 1} - 1 \right| &= \left| \frac{x^2 - \cos x - (x^2 + 1)}{x^2 + 1} \right| = \left| \frac{-\cos x - 1}{x^2 + 1} \right| = \left| \frac{\cos x + 1}{x^2 + 1} \right| \\ &\leq \frac{2}{x^2 + 1} \\ &\leq \frac{2}{x^2} \end{aligned}$$

$$\Rightarrow |f(x) - 1| < \epsilon \quad \text{if} \quad \frac{2}{x^2} < \epsilon$$

$$\therefore \frac{2}{x^2} < \epsilon \Leftrightarrow \sqrt{\frac{2}{\epsilon}} < x \quad \text{for } x > 0, \quad M \triangleq \sqrt{\frac{2}{\epsilon}}$$

2.4 Proofs of basic limit results

Not assessable.

2.5 Limits of functions at a point

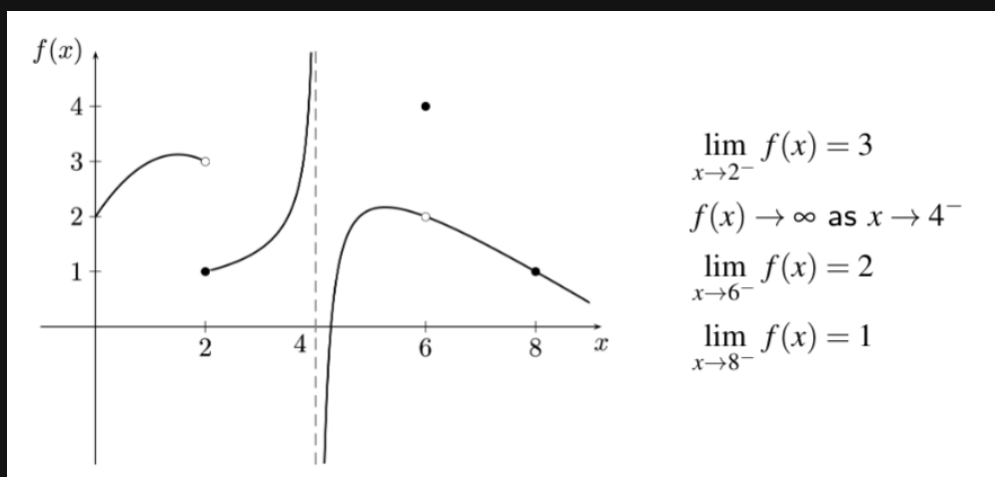
2.5.1 Left-hand, right-hand and two-sided limits

- If $f(x)$ approaches L as x approaches a from the left-hand side, then:

$$\lim_{x \rightarrow a^-} f(x) = L \quad (\text{left-hand limit})$$

- If $f(x)$ gets arbitrarily large as x approaches a from the left side, then:

$$f(x) \rightarrow \infty \quad \text{as } x \rightarrow a^-$$

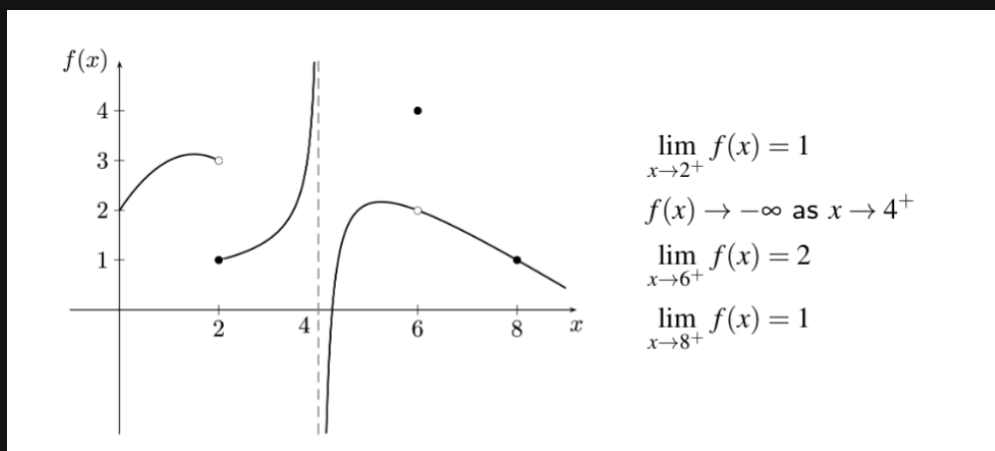


- If $f(x)$ approaches L as x approaches a from the right-hand side, then:

$$\lim_{x \rightarrow a^+} f(x) = L \quad (\text{right-hand limit})$$

- If $f(x)$ gets arbitrarily large as x approaches a from the right side, then:

$$f(x) \rightarrow \infty \quad \text{as } x \rightarrow a^+$$



- Left I be an open interval that contains the point a . For a real-valued function f defined on I except possibly at a , if both one-sided limits exist and

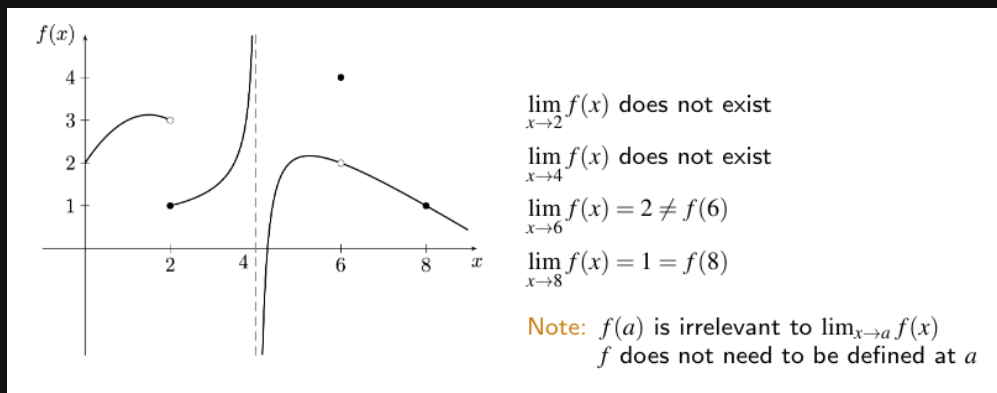
$$\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = L,$$

then we say that the (two-sided) limit of $f(x)$ as $x \rightarrow a$ exists and is equal to L , and we write

$$\lim_{x \rightarrow a} f(x) = L.$$

Otherwise, $\lim_{x \rightarrow a} f(x)$ does not exist.

- Note:** $f(a)$ is irrelevant to $\lim_{x \rightarrow a} f(x)$. f does not need to be defined at a



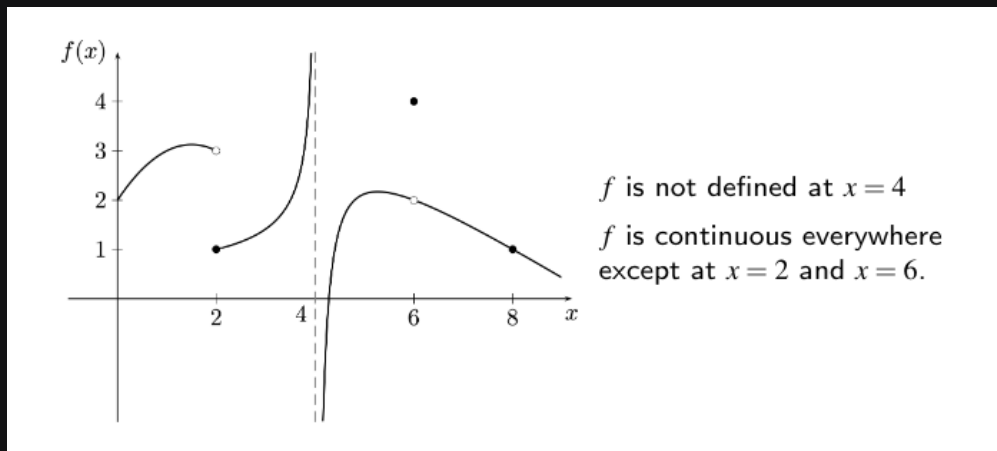
2.5.2 Limits and continuous functions

Let I be an open interval that contains the point a . A function $f : I \rightarrow \mathbb{R}$ is continuous at a if

$$\lim_{x \rightarrow a} f(x) = f(a)$$

Otherwise, we say that f is discontinuous at a .

If f is continuous at every point of $\text{Dom}(f)$, then we say that f is continuous everywhere.



2.5.3 Rules for limits at a point

Arithmetic

Suppose that $a \in \mathbb{R}$, and $\lim_{x \rightarrow a} f(x) = L_1$ and $\lim_{x \rightarrow a} g(x) = L_2$. Then

- $\lim_{x \rightarrow a} (f(x) + g(x)) = L_1 + L_2$
- $\lim_{x \rightarrow a} (f(x) - g(x)) = L_1 - L_2$
- $\lim_{x \rightarrow a} f(x)g(x) = L_1L_2$
- $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{L_1}{L_2}$

Function composition

If $\lim_{x \rightarrow a} f(x) = L$ and g is continuous at L , then

$$\lim_{x \rightarrow a} g(f(x)) = g\left(\lim_{x \rightarrow a} f(x)\right) = g(L)$$

Example

Find the limit $\lim_{x \rightarrow 5} \sqrt{2x + \cos \pi x^2}$.

$$\begin{aligned} \lim_{x \rightarrow 5} \sqrt{2x + \cos \pi x^2} &= \sqrt{\lim_{x \rightarrow 5} 2x + \cos \pi x^2} \quad (\sqrt{} \text{ is continuous}) \\ &= \sqrt{\lim_{x \rightarrow 5} 2x + \lim_{x \rightarrow 5} \cos \pi x^2} \\ &= \sqrt{\lim_{x \rightarrow 5} 2x + \cos \lim_{x \rightarrow 5} \pi x^2} \quad (\cos \text{ is continuous}) \\ &= \sqrt{10 + \cos 5^2 \pi} \quad (\cos 5^2 \pi = -1) \\ &= 3 \end{aligned}$$

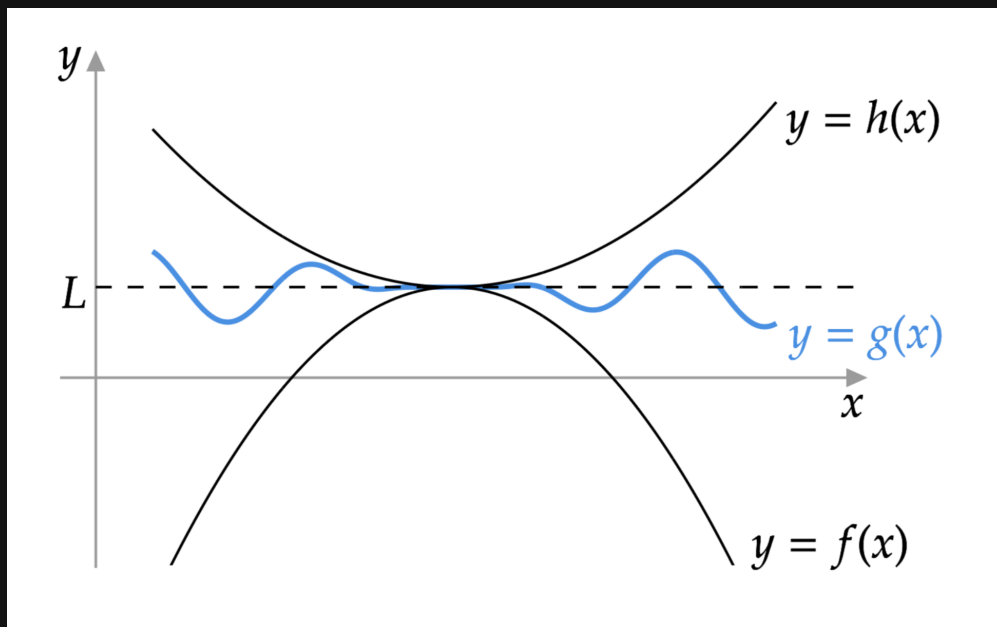
Pinching theorem (two-sided limit)

Let I be an open interval that contains the point a . Suppose that f , g , and h are functions defined on I . If

- $f(x) \leq g(x) \leq h(x), \quad \forall x \in I$, and
- $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x) = L$

then

$$\lim_{x \rightarrow a} g(x) = L$$



Example

Suppose that $f : \mathbb{R} \rightarrow \mathbb{R}$ is defined by the formula

$$f(x) = \begin{cases} \frac{|x^2-9|}{x-3} & \text{if } x \neq 3 \\ -6 & \text{if } x = 3 \end{cases}$$

Discuss the limiting behaviour of $f(x)$ as x approaches 3.

Limit from left hand side

$$\begin{aligned} \text{For } x \in (-3, 3), \quad \frac{|x^2-9|}{x-3} &= \frac{|x+3||x-3|}{x-3} = -(x+3) \\ \Rightarrow \lim_{x \rightarrow 3^-} f(x) &= \lim_{x \rightarrow 3^-} -(x+3) = -6 = f(3) \end{aligned}$$

Limit from right hand side

$$\begin{aligned} \text{For } x > 3, \quad \frac{|x^2-9|}{x-3} &= x+3 \\ \Rightarrow \lim_{x \rightarrow 3^+} f(x) &= \lim_{x \rightarrow 3^+} x+3 = 6 \end{aligned}$$

Since $\lim_{x \rightarrow 3^+} f(x) \neq \lim_{x \rightarrow 3^-} f(x)$, two-sided limit does not exist at $x = 3$

So, f is not continuous at $x = 3$

Chapter 3

Properties of continuous functions

3.1 Combining continuous functions

Suppose that the functions f and g are continuous at a point a . Then

1. $f + g$, $f - g$ and fg are continuous at a
2. $\frac{f}{g}$ is continuous at a , provided that $g(a) \neq 0$

$$\begin{aligned}\lim_{x \rightarrow a} (f + g)(x) &= \lim_{x \rightarrow a} (f(x) + g(x)) && \text{(by definition of } f + g) \\ &= \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x) && \text{(by "limit of sum = sum of limits")} \\ &= f(a) + g(a) && \text{(by continuity of } f \text{ and } g \text{ at } a) \\ &= (f + g)(a)\end{aligned}$$

Suppose that the function f is continuous at a , and g is continuous at $f(a)$. Then $g \circ f$ is continuous at a .

$$\begin{aligned}\lim_{x \rightarrow a} (g \circ f)(x) &= \lim_{x \rightarrow a} g(f(x)) \\ &= g\left(\lim_{x \rightarrow a} f(x)\right) && \text{(by limit rule of function composition)} \\ &= g(f(a)) && \text{(by continuity of } f \text{ at } a) \\ &= (g \circ f)(a)\end{aligned}$$

Example

Every polynomial $p : \mathbb{R} \rightarrow \mathbb{R}$ is continuous at every $a \in \mathbb{R}$.

$$\begin{aligned}\Leftarrow \text{for any } n \in \mathbb{N}, \quad p(x) &= a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0 \\ &= \text{product \& sum of } g(x) = x \text{ and constant functions, ie. continuous by (i)}\end{aligned}$$

3.2 Continuity on intervals

For a real-valued function f defined on an open interval (a, b) , we say that f is continuous on (a, b) if it is continuous at every $x \in (a, b)$.

For a real-valued function f defined on a closed interval $[a, b]$, we say that

- f is continuous at the endpoint a if $\lim_{x \rightarrow a^+} f(x) = f(a)$
- f is continuous at the endpoint b if $\lim_{x \rightarrow b^-} f(x) = f(b)$
- f is continuous on $[a, b]$ if it is continuous on (a, b) and at endpoints a and b .

3.3 The intermediate value theorem

Suppose that f is continuous on $[a, b]$, and $f(b) < z < f(a)$ for some z . The graph of f from $(a, f(a))$ to $(b, f(b))$ without lifting the pen. We must cross a point (c, z) along the way for some $c \in (a, b)$.

Let $f : [a, b] \rightarrow \mathbb{R}$ be a continuous function. If either $f(a) < z < f(b)$ or $f(b) < z < f(a)$ for some $z \in \mathbb{R}$, then there is at least one $c \in (a, b)$ such that $f(c) = z$

- In general, c is not unique
- The continuity of f is necessary
- Only for continuous functions defined on (subsets of) $\mathbb{R}$, no analogue for $\mathbb{Q}$

$$\begin{array}{ll} f : \mathbb{R} \rightarrow \mathbb{R}, & f(x) = x^2 - 2 \\ f \text{ is continuous on } \text{Dom}(f) = \mathbb{R} \\ & = \in (f(0), f(2)), f(c) = 0 \text{ at } c = \sqrt{2} \in (0, 2) \\ f : \mathbb{Q} \rightarrow \mathbb{Q}, & g(x) = x^2 - 2 \\ g \text{ is continuous on } \text{Dom}(g) = \mathbb{Q} \\ & = \in (g(0), g(2)), g(c) = 0 \text{ at } c = \sqrt{2} \notin \mathbb{Q} \end{array}$$

- Since infinity is not part of $\mathbb{R}$, a definite bound must be defined before using IVT

Example

Show that there exists a solution $c \in [1, 2]$ of the equation $\sqrt{c} = c^2 - 1$ and approximate its value.

Define a function $f : [1, 2] \rightarrow \mathbb{R}$ by $f(x) = x^2 - 1 - \sqrt{x}$. Then,

- f is continuous on $[1, 2]$
- $f(1) = -1$ and $f(2) = 3 - \sqrt{2} > 0$.

Since $f(1) < 0 < f(2)$, by **IVT**, there is a number $c \in (1, 2)$ such that $f(c) = c^2 - 1 - \sqrt{c} = 0$

Approximation:

- take $c_1 = 1.5 \Rightarrow f(c_1) = 0.0253$, by IVT, $\exists$ a solution $c \in (1, c_1)$
- take $c_2 = 1.25 \Rightarrow f(c_2) = -0.5555$, by IVT, $\exists$ a solution $c \in (1, c_2)$

3.4 The maximum-minimum theorem

Let f be a real-valued function defined on a closed interval $[a, b]$.

- A point $c \in [a, b]$ is a global minimum point for f on $[a, b]$ if

$$f(c) \leq f(x), \quad \forall x \in [a, b].$$

- A point $d \in [a, b]$ is a global maximum point for f on $[a, b]$ if

$$f(d) \geq f(x), \quad \forall x \in [a, b].$$

If f is continuous on $[a, b]$, then f attains its minimum and maximum on $[a, b]$, ie. there exists points $c, d \in [a, b]$ such that

$$f(c) \leq f(x) \leq f(d), \quad \forall x \in [a, b].$$

- Closed intervals are necessary
- Continuity of f is necessary

Example

Let f be a function defined by $f(x) = \begin{cases} \frac{1}{x^2+1} \sin \frac{1}{x}, & \text{if } x \neq 0 \\ 1, & \text{if } x = 0 \end{cases}$

1. Is f bounded on $[-2, 2]$?

$$\text{Yes. For } x \neq 0, \quad |f(x)| = \left| \frac{1}{x^2+1} \sin \frac{1}{x} \right| \leq \frac{1}{x^2} \leq 1.$$

$$\text{For } x = 0, \quad f(x) = 1.$$

$$\Rightarrow |f(x)| \leq 1, \quad \forall x \in [-2, 2]$$

2. Does it attain a minimum or maximum on $[-2, 2]$

- f on $[-2, 2]$ attains a maximum at $x = 0$.
- f does not have a minimum value on $[-2, 2]$

$$\text{For } x_n = \left(\frac{-\pi}{2} + 2n\pi \right)^{-1}, \quad n \in \mathbb{Z}$$

$$\sin \frac{1}{x_n} = -1$$

$$f(x_n) = -\frac{1}{\left(\frac{-\pi}{2} + 2n\pi \right)^{-2} + 1}$$

Then as $n \rightarrow \infty$, $x_n \rightarrow 0$ and $f(x_n) \rightarrow -1$

Note: $f(x) > -1$

$\Rightarrow$ as $x \rightarrow 0$, $f(x)$ oscillates between (not touching) -1 and 1

Chapter 4

Differentiable functions

4.1 Gradients of tangents and derivatives

Suppose that f is defined on some open interval containing the point x . It is said that f is differentiable at x if

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \text{ exists}$$

This limit is denoted by $f'(x)$, $\frac{df}{dx}(x)$ or $\frac{d}{dx}f(x)$ and is called the derivative of f at x .

Example

1. Show that $\lim_{h \rightarrow 0} \frac{\cos(h) - 1}{h} = 0$. (Hint: $\cos(h) - 1 = \frac{\cos^2(h) - 1}{\cos(h) + 1} = -\frac{\sin^2(h)}{\cos(h) + 1}$)

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{\cos(h) - 1}{h} &= -\lim_{h \rightarrow 0} \frac{\sin(h)}{\cos(h) + 1} \frac{\sin(h)}{h} \\ &= -\frac{\lim_{h \rightarrow 0} \sin(h)}{\lim_{h \rightarrow 0} \cos(h) + 1} \lim_{h \rightarrow 0} \frac{\sin(h)}{h} \quad \left(\lim_{h \rightarrow 0} \frac{\sin(h)}{h} = 1 \right) \\ &= -\frac{0}{2} \times 1 = 0 \end{aligned}$$

2. For which $n \in \mathbb{N}$ is the function f defined by

$$f(x) = \begin{cases} x^n \sin\left(\frac{1}{x}\right) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

continuous and differentiable at $x = 0$

For continuity, $\lim_{x \rightarrow 0} x^n \sin\left(\frac{1}{x}\right) = f(0) = 0$.

$$1 \leq \sin\left(\frac{1}{x}\right) \leq 1, \forall x \neq 0 \Rightarrow -|x^n| \leq x^n \sin\left(\frac{1}{x}\right) \leq |x^n|, \forall x \neq 0, \forall n \in \mathbb{N}$$

$$(\text{Pinching theorem}) \Rightarrow \lim_{x \rightarrow 0} x^n \sin\left(\frac{1}{x}\right) = 0, \text{ ie. continuous at } x = 0, \forall n \in \mathbb{N} \setminus \{0\}$$

For differentiability, $\lim_{h \rightarrow 0} \frac{1}{h} [f(0+h) - f(0)]$ exists.

$$\begin{aligned} &= \frac{1}{h} \left[h^n \sin\left(\frac{1}{h}\right) - 0 \right] \\ &= h^{n-1} \sin\left(\frac{1}{h}\right) \rightarrow 0, \quad \forall n-1 \in \mathbb{N} \setminus \{0\} \end{aligned}$$

ie. $n \in \mathbb{N}, n \neq 0, 1$

4.2 Rules for differentiation

Suppose that f and g are differentiable at x . Then $f + g$, $f - g$ and fg are differentiable at x , and $\frac{f}{g}$ is also differentiable at x , provided $g(x) \neq 0$.

1. $(f + g)'(x) = f'(x) + g'(x)$
2. $(Cf)'(x) = Cf'(x)$ where C is any constant
3. $(fg)'(x) = f'(x)g(x) + f(x)g'(x)$, (product rule)
4. $\left(\frac{f}{g}\right)'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2}$, provided that $g(x) \neq 0$ (quotient rule)

Suppose that g is differentiable at x , and f is differentiable at the point $g(x)$. Then $f \circ g$ is differentiable at x and

$$(f \circ g)'(x) = f'(g(x))g'(x)$$

4.3 Proofs of results

lol no

4.4 Implicit differentiation

A function defined by an equation relating x and y is implicitly defined.

Example

Determine the tangent to the curve $x^4 - x^2y^2 + y^4 = 13$ at $(2, 1)$

y depends on $x : y(x)$

in a neighbourhood of $(2, 1)$

Differentiate both sides of the equation with respect to x :

$$4x^3 - 2xy^2(x) - 2x^2y(x)y'(x) + 4y^3(x)y'(x) = 0$$

At $(x, y) = (2, 1)$,

$$32 - 4 - 8y'(2) + 4y'(2) = 0$$

$$28 - 4y'(2) = 0$$

$$y'(2) = 7$$

Tangent line at $(2, 1)$:

$$y - 1 = 7(x - 2)$$

4.5 Differentiation, continuity and split functions

- If f is differentiable at a , then f is continuous at a
- If f is not continuous at a , then it is not differentiable at a
- If f is continuous at a , it may not be differentiable at a

– eg. $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = |x|$ and $a = 0$

Suppose $a \in \mathbb{R}$ and the function f is defined by

$$f(x) = \begin{cases} p(x) & \text{if } x \geq a \\ q(x) & \text{if } x < a \end{cases}$$

where p and q are differentiable in some open interval containing a . If

- f is continuous at a ($\Leftrightarrow p(a) = q(a)$) and
- $p'(a) = q'(a)$,

then f is differentiable at $x = a$.

4.6 Derivatives and function approximation

If f is differentiable at a , then

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

By the definition of a limit, $\frac{f(x) - f(a)}{x - a}$ gets close to $f'(a)$ as x gets close to a . This statement is equivalent to

$$\begin{aligned}\frac{f(x) - f(a)}{x - a} &\approx f'(a) \\ f(x) &\approx f(a) + f'(a)(x - a)\end{aligned}$$

Example

Estimate $\sin(0.51\pi)$

$$\begin{aligned}\sin(0.51\pi) &\approx \sin(0.5\pi) + \cos(0.5\pi)(0.51\pi - 0.5\pi) \\ &= 1 + 0 \times (0.01\pi) \\ &= 1 \\ \Rightarrow \sin(0.51\pi) &\approx 1\end{aligned}$$

4.7 Derivatives and rates of change

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4.8 Local maximum, local minimum and stationary points

Let f be defined on some interval I .

- A point $c \in I$ is a local minimum if there exists a $h > 0$ such that

$$f(x) \geq f(c), x \in (c - h, c + h) \cap I$$

- A point $d \in I$ is a local maximum point if there exists a $h > 0$ such that

$$f(x) \leq f(d), x \in (d - h, d + h) \cap I$$

Chapter 5

The mean value theorem and applications

5.1 The mean value theorem

Suppose that f is

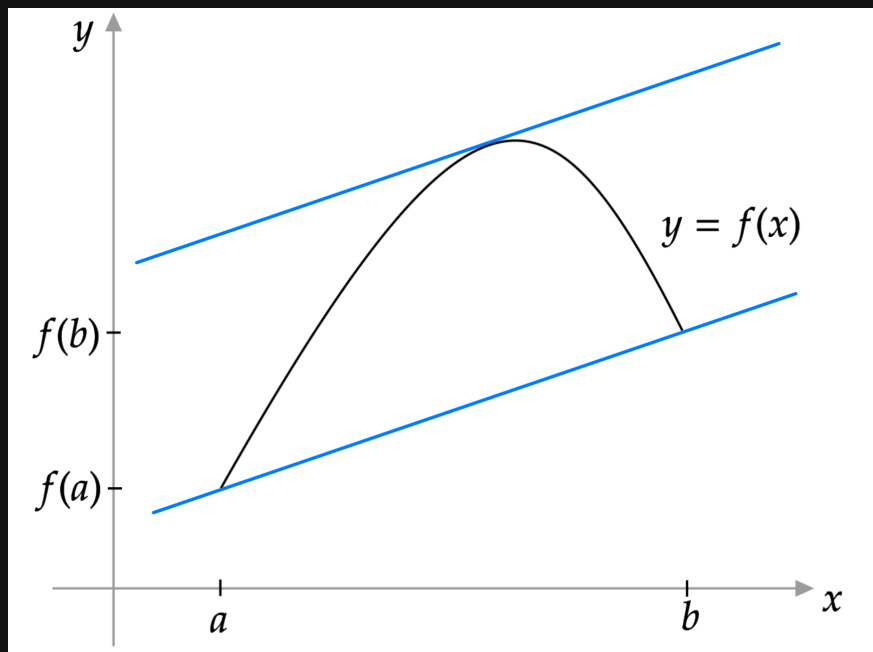
1. continuous on $[a, b]$, and
2. differentiable on (a, b)

Then there is at least one $c \in (a, b)$ such that

$$\frac{f(b) - f(a)}{b - a} = f'(c)$$

The mean value theorem cannot be applied if

- f is not continuous on $[a, b]$
- f is not differentiable on (a, b)



Example

A car enters a tunnel at a speed of 30km/h and after 1 minute leaves the tunnel at a speed of 40km/h. The length of the tunnel is 1km. Did the driver break the speed limit of 50km/h?

Let $x : [t_{in}, t_{out}] \rightarrow \mathbb{R}$, $x(t)$ = position of the car at time t

Assume x is continuous on $[t_{in}, t_{out}]$ and differentiable on (t_{in}, t_{out})

$\Rightarrow$ By MVT, $\exists T \in (t_{in}, t_{out})$ such that

$$x'(t) = \frac{x(t_{out}) - x(t_{in})}{t_{out} - t_{in}} = \frac{1}{\frac{1}{60}} \text{ km/h} = 60 \text{ km/h}$$

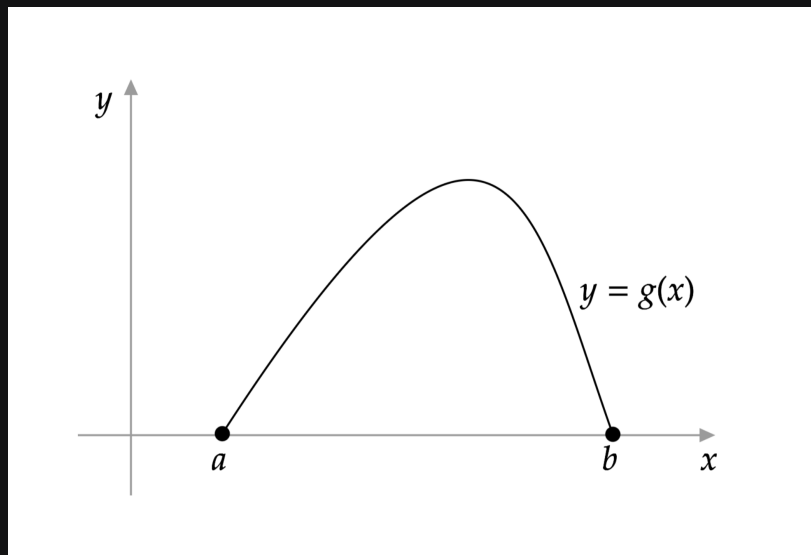
Therefore, the driver broke the speed limit

5.2 Proof of the mean value theorem

Suppose that g

- is continuous on $[a, b]$
- differentiable on (a, b)
- $g(a) = g(b) = 0$ (optional)

Then there is $c \in (a, b)$ such that $g'(c) = 0$



Case 1: $g(x) = 0, \forall x \in [a, b]$
 $\Rightarrow g'(c) = 0$ for any $c \in (a, b)$

Case 2: $g(d) > 0$ for some $d \in (a, b)$
 By continuity and MMT, g attains a maximum value at some $c \in [a, b]$

Then $g(c) \geq g(d) > 0 = g(a) = g(b)$
 $\Rightarrow c \neq a, c \neq b$, so $c \in (a, b)$
 $\Rightarrow g'(c) = 0$

Case 3: $g(x) \leq 0, \forall x \in [a, b]$, and g is not a constant on $[a, b]$
 Since $g(a) = g(b)$, Case 3 $\Leftrightarrow g(d) < 0$ for some $d \in (a, b)$
 Similar to Case 2, g attains a minimum value at some $c \in (a, b)$, $g'(c) = 0$

To prove MVT, a function f is "deformed" such that it looks like g in Rolle's theorem.

Define $g : [a, b] \rightarrow \mathbb{R}$ by

$$g(x) = f(x) - \frac{f(b) - f(a)}{b - a}x$$

Hence

$$\exists c \in (a, b) \text{ such that } g'(c) = 0 \quad \Leftrightarrow \quad f'(c) - \frac{f(b) - f(a)}{b - a} = 0$$

5.3 Proving inequalities using the mean value theorem

From the equality $\frac{f(b) - f(a)}{b - a} = f'(c)$ in MVT,

$$m \leq f'(c) \leq M \quad \Leftrightarrow \quad m \leq \frac{f(b) - f(a)}{b - a} \leq M$$

Example

Use MVT to show that $e^x > 1 + x$ for all $x > 0$

For any $x > 0$, define a function

$$f : [0, x] \rightarrow \mathbb{R}, \quad f(t) = e^t$$

- f is continuous on $[0, x]$
- f is differentiable on $(0, x)$

Hence, by MVT, $\exists c \in (0, x)$ such that $\frac{f(x) - f(0)}{x - 0} = f'(c)$

$$\Updownarrow$$

$$\frac{e^x - 1}{x} = e^c > 1$$

$$e^x - 1 > x$$

$$e^x > 1 + x$$

Example

Use MVT to show that $\ln(x) < x - 1$ for all $x > 1$

For any $x > 1$, define a function

$$f : [1, x] \rightarrow \mathbb{R}, \quad f(t) = \ln(t)$$

By MVT, $\exists c \in (1, x)$ such that $\frac{f(x) - f(1)}{x - 1} = f'(c)$

$$\Updownarrow$$

$$\frac{\ln(x) - 0}{x - 1} = \frac{1}{c} < 1$$

$$\ln(x) < x - 1$$

5.4 Error bounds

From the equality $\frac{f(b) - f(a)}{b - a} = f'(c)$ in MVT,

$$|f(b) - f(a)| \leq M \iff |f'(c)| |b - a| \leq M$$

Example

How good an approximation of $\cos 1$ is $\cos \frac{\pi}{3}$

By MVT, $\exists c \in (1, \frac{\pi}{3})$ such that

$$\frac{\cos(1) - \cos\left(\frac{\pi}{3}\right)}{1 - \frac{\pi}{3}} = f'(c) = -\sin(c)$$

$$\begin{aligned}\Rightarrow 0 < \cos(1) - \cos\left(\frac{\pi}{3}\right) &= -\sin(c) \left(1 - \frac{\pi}{3}\right) \\ &= \sin(c) \left(\frac{\pi}{3} - 1\right)\end{aligned}$$

$$\Rightarrow 0.5 < \cos(1) < 0.55$$